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Adaptive Multi-Feature Fusion-Based UAV Detection and Tracking Collaboration Framework Trigger Mechanism

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***Abstract*—The popularization of unmanned aerial vehicle (UAV) technology has brought challenges to the security of sensitive areas and low-altitude traffic management. Unauthorized UAV intrusions may pose security threats. However, existing vision-based Anti-UAV solutions have significant drawbacks: their trigger mechanisms rely on fixed thresholds or single features, failing to adapt to dynamic scenarios such as infrared low-illumination and dense occlusion; meanwhile, multi-feature fusion methods either ignore motion continuity or lack dynamic thresholds, resulting in poor cross-modal adaptability. To address these issues, this study proposes an Adaptive Multi-Feature Fusion Trigger Mechanism (AMFT) and constructs a "tracking-led, detection-on-demand triggering" detection-tracking collaboration framework. The core components of this mechanism include: 1) a three-dimensional evaluation system of "confidence-mobility-interference degree" to quantify target matching degree, motion smoothness, and background discrimination; 2) a dynamic threshold module based on sliding window scene complexity to adjust detection trigger conditions in real time. Extensive experiments verify the superiority of the proposed AMFT strategy, as well as the necessity of multi-feature fusion and dynamic thresholds. This method breaks through the limitations of traditional single trigger mechanisms and improves the robustness and generalization ability of Anti-UAV systems in complex scenarios.**

Keywords—Anti-UAV; Detection-Tracking Collaboration; Adaptive Trigger Mechanism; Multi-Feature Fusion; Dynamic Threshold

I. INTRODUCTION

The rapid popularization of UAV technology has brought convenience to various fields, while also posing severe challenges to the security of sensitive areas and low-altitude traffic management. Unauthorized UAV intrusions may lead to privacy leaks, facility damage, and even national security threats [1]. As the core technology to address such threats, Anti-Unmanned Aerial Vehicle (Anti-UAV) systems require their perception layer to achieve real-time positioning and continuous tracking of UAVs. Visual perception has become the preferred solution for both military and civilian scenarios due to its advantages of low cost and flexible deployment [2]. According to the 2024 Report by the International Civil Aviation Organization (ICAO), the number of incidents involving unauthorized UAVs interfering with airport operations has increased significantly worldwide, and most of these incidents resulted in ineffective disposal due to response delays or false alarms of existing systems [3]. More critically, coordinated intrusions by UAV swarms have significantly elevated the threat level, and existing systems suffer from insufficient accuracy in simultaneously tracking multiple UAVs [4].

Existing non-visual Anti-UAV technologies have obvious limitations: radar systems have low detection rates for micro-UAVs and high costs [5]; radio frequency detection is susceptible to electromagnetic interference and has a high false alarm rate [6]; acoustic sensors exhibit a sharp decline in recognition accuracy in complex environments [7]. In contrast, computer vision-based solutions are more cost-effective and achieve higher recognition rates for micro-UAVs [8], but they face bottlenecks in three typical scenarios: (1) Infrared low-illumination scenarios: low target resolution and missing appearance features lead to reduced recognition rates [6]; (2) Dense occlusion scenarios: UAVs are easily occluded, resulting in prolonged tracking loss [9]; (3) UAV swarm scenarios: mutual occlusion between targets causes identity (ID) switching issues [4].

Most existing vision-based Anti-UAV solutions adopt a "global detection + local tracking" collaboration mode [10-12]. However, the trigger timing of global detection after tracking loss directly determines system performance. Premature triggering leads to redundant computations, while delayed triggering prolongs the target loss duration [13-14]. Current mainstream trigger mechanisms have two major defects: first, fixed threshold strategies cannot adapt to dynamic scenarios − for example, the confidence threshold method proposed by Wang et al. performs poorly in scenarios with fast motion and background interference [14]; second, single-feature evaluation lacks comprehensiveness − for instance, the IoU-based trigger strategy proposed by Jiang et al. has a high false trigger rate in scenarios where UAV scale changes [15]. More importantly, most existing methods are based on

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simplified datasets, which differ significantly from target features in real-world scenarios [30].

Although recent attempts at multi-feature fusion have made progress, shortcomings remain: the Evidential Detection-Tracking Collaboration (EDTC) framework proposed by Zhu et al. fuses features via evidence theory but fails to consider motion continuity [17]; the Siamese Dual Fusion (SiamDT) framework proposed by Zhang et al. improves modal adaptability but does not design dynamic thresholds [18]. Additionally, the problem of poor cross-modal scenario adaptability is prominent—the optimal model for visible light scenarios exhibits a significant performance decline in infrared scenarios [6], restricting the all-weather operation of the system.

This study proposes an Adaptive Multi-Feature Fusion Trigger Mechanism (AMFT), with the following core contributions:

1. Construct a three-dimensional evaluation system of "confidence-mobility-interference degree" to quantify background interference and motion smoothness;
2. Design a scene-adaptive threshold module to dynamically adjust trigger conditions based on sliding window complexity;
3. Verify on multi-modal datasets that the comprehensive performance is significantly improved, and the real-time performance meets practical requirements.

II. Related Work

Three typical technical modes exist in the field of target detection and tracking, with distinct core design logics and performance characteristics:

A. Detection-led mode

High-frequency detection serves as the core, and tracking is only used for inter-frame association. For example, the YOLOX+DeepSort solution runs detection on every frame; although it performs well on datasets, its inference speed is slow [21]. Cascade-RCNN improves the detection rate of small targets through multi-stage regression but cannot be deployed on edge devices due to its large parameter size [6,22]. The fundamental defect of such solutions lies in computational redundancy [23].

B. Tracking-led mode:

Tracking is the core, and detection is triggered at low frequency. SiamRPN++ achieves real-time tracking via a siamese network but has a high tracking loss rate in small-target scenarios [24]. TransT introduces the Transformer attention mechanism and performs well on datasets, but its large parameter size results in insufficient speed on edge devices [25]. The optimization of trigger strategies is the research focus of this mode; existing methods still suffer from invalid detection or false triggers in scenarios with fast motion and dynamic backgrounds [14,26].

C. Hybrid collaboration mode:

It jointly optimizes detection and tracking through an end-to-end approach. DiMP improves robustness using a discriminative model but lacks a re-detection mechanism [27]; the EDTC framework proposes an evidence theory-based switching strategy but ignores the impact of scene complexity on thresholds [17]; SiamDT fuses dual-modal features to enhance cross-modal adaptability but fails to solve the trigger timing problem [18].

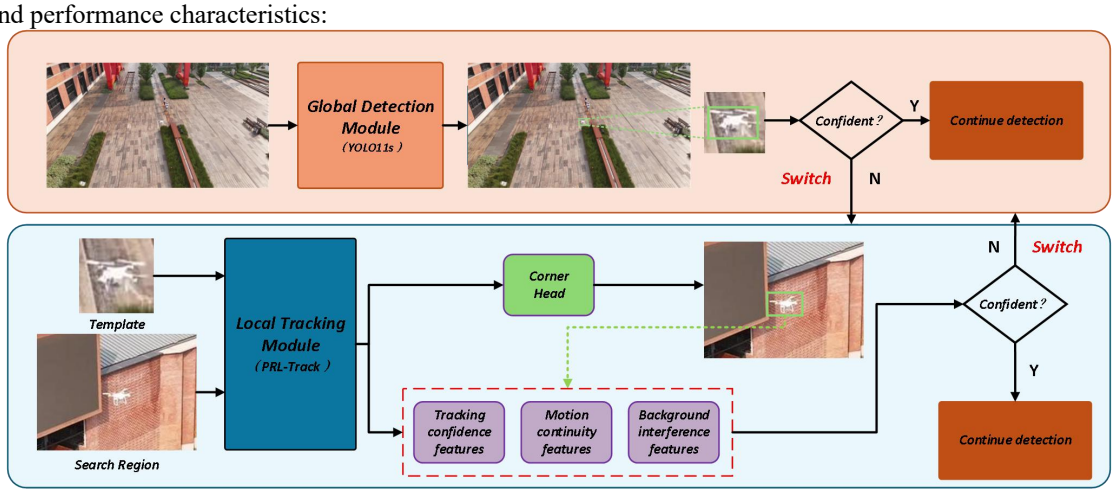


Fig. 1. The pipeline of the proposed AMFT approach.

III. Method

A. Overall Framework Design

The proposed detection-tracking collaboration framework adopts the core logic of "tracking-led, detection-on-demand triggering". It achieves a balance between real-time performance and robustness through close collaboration among the global detection module, local tracking module, and adaptive trigger decision module, adapting to the dynamic requirements of Anti-

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UAV scenarios (as shown in Fig. 1). The input video stream first enters the local tracking module for continuous target tracking. In the initial frame or scenarios requiring detection triggering, the global detection module first acquires the initial target position and initializes the tracker for the local tracking module, providing a basic positioning basis for subsequent tracking. During the tracking process, the adaptive trigger decision module, as the core connection component, fuses multi-dimensional features in real time to evaluate the current tracking reliability. If the evaluation result meets the trigger conditions, it immediately activates the global detection module to reposition the target, and feeds back the updated target position to the local tracking module to correct the tracking state and avoid tracking loss. When the tracking state is stable, the adaptive trigger decision module maintains the local tracking mode to reduce redundant calls to the global detection module. Finally, the local tracking module outputs the target bounding box for each frame. If the global detection module fails to capture the target multiple times, it synchronously outputs a "no target" identifier. The three modules form an efficient collaborative detection-tracking system through closed-loop linkage of "initialization-tracking evaluation-on-demand detection-state update".

B. Adaptive Multi-Feature Fusion Trigger Mechanism (AMFT)

This module consists of three sub-steps: multi-dimensional reliability feature extraction, comprehensive reliability score calculation, and dynamic threshold update. The detailed design is as follows:

- Tracking confidence features S_{conf} : Directly obtain the target confidence score output from the tracker's classification branch, normalized to the [0,1] range. S_{conf} reflects the model's intrinsic confidence in the current tracking result; values closer to 1 indicate higher feature matching accuracy with the template and more reliable tracking.
- Motion continuity feature S_{mot} : The reliability is judged by combining the smoothness of target motion to avoid false detection triggers caused by "instantaneous confidence fluctuations". The calculation method is as follows. IoU_t represents the IoU value between the current frame's tracking box and the previous frame's tracking box (measuring positional continuity); D_t denotes the deviation rate between the current frame's target center displacement and the average displacement of the preceding three frames (measuring motion plausibility). A higher deviation rate indicates more unstable motion (e.g., sharp turns, sudden occlusions). Finally, S_{mot} is normalized to the range [0,1], where higher values indicate more continuous motion.

$$S_{mot} = 0.6 \times IoU_t + 0.4 \times (1 - D_t) \quad (1)$$

- Background interference feature S_{bg} : For "target-background fusion," the distinguishability between targets and backgrounds is evaluated through feature similarity: HOG features are extracted from the tracking box region and the surrounding background area three times its size. The cosine similarity A between the two feature sets is calculated, then normalized to [0,1] as B. Higher values indicate stronger target-background distinguishability and reduced interference.

A weighted fusion strategy is finally adopted to calculate the comprehensive reliability score of each frame:

$$R_t = \omega_1 \times S_{conf} + \omega_2 \times S_{mot} + \omega_3 \times S_{bg} \quad (2)$$

C. Dynamic Threshold Update Strategy

Traditional fixed thresholds cannot adapt to changes in scene complexity. This paper proposes a sliding-window-based dynamic threshold τ_t . The strategy for updating via dynamic thresholds involves three steps: First, configure a sliding window of length $K = 10$ to store the composite reliability scores $\{R_{t-9}, \ldots, R_t\}$ of the most recent 10 frames, ensuring coverage of "short-term scene changes"; Second, calculate scene complexity. The standard deviation σ_t of scores within the window reflects scene dynamics; a larger σ_t indicates greater scene complexity (e.g., sudden occlusions, rapid motion).

Finally, calculate threshold $\tau_t = \tau_0 - \alpha \times \sigma_t$, where $\tau_0 = 0.7$ (base threshold, optimized via the validation set) and $\alpha = 0.5$ (adjustment coefficient) — the more complex the scene (large σ_t), the lower the threshold, triggering detection earlier to prevent tracking loss; when the scene is stable (small σ_t), the higher the threshold, reducing false detections.

D. Trigger Logic and Collaboration Strategy

The triggering logic and collaborative strategy for target detection and tracking can be divided into four core stages according to the workflow, with each stage seamlessly integrated to achieve efficient and precise target monitoring: During the initial stage, the system first performs global detection on the first 5 frames of the video stream. If a target is successfully detected, the tracker is initialized using the detection box; if no target is detected, it is directly marked as "Target Not Present". Upon entering the tracking phase, the system calculates tracking reliability parameter R_t and decision threshold τ_t for each frame. When $R_t < \tau_t$ occurs, the current tracking state is deemed unreliable, triggering a global detection to relocate the target. During the detection feedback loop, if the target is successfully captured after

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triggering detection, the system updates the tracker's template parameters and target position using the new detection box to ensure tracking accuracy. If the target remains undetected, it is marked as " Target Disappeared," and a persistent detection mechanism is initiated (performing a global detection every 3 frames) until the target reappears. To optimize resource utilization, the system must immediately revert to tracking mode in the frame following a global detection trigger. This prevents redundant computational resources caused by consecutive detection operations, balancing detection accuracy with operational efficiency.

IV. EXPERIMENTAL

A. Experimental Settings

- Dataset Details: The method in this paper uses the DUT Anti-UAV dataset [9] as the core validation set, which includes a detection subset and a tracking subset. For the detection subset, it includes a training set (5200 images), a validation set (2600 images) and a test set (2200 images), which are mainly used to train the target detector and optimize the trigger mechanism parameters of detection-tracking in the framework to improve the rationality and accuracy of the trigger logic; for the tracking subset, it covers 20 video sequences (a total of 24804 frames), which can be used to evaluate the stability and robustness of the framework's detection-tracking collaboration under complex working conditions.

- Metrics: In traditional UAV detection-tracking tasks, system performance evaluation usually adopts indicators that take into account accuracy, stability and efficiency. To comprehensively evaluate the performance of the proposed UAV detection-tracking collaboration framework, this study selects three core indicators: tracking stability, positioning accuracy and real-time performance, combining scene characteristics and task requirements [8][9].

- Baselines: To verify the effectiveness of the trigger mechanism of the proposed framework, three types of baselines are selected, and tests are conducted under the same model/environment/dataset to ensure fairness: FT (DUT benchmark, threshold 0.9) is simple and low-cost, but cannot adapt to scene dynamics; IT (triggered when IoU<0.5) reflects spatial correlation, but ignores target appearance changes; RT (triggered when SiamFC peak < 0.6) reflects internal confidence, but has limited robustness to fast motion/occlusion.

- Implement Details: All experiments (model training, performance testing) in this paper are completed in a unified hardware and software environment. We use the PyTorch framework to implement the model proposed in this paper, and use two NVIDIA GeForce RTX 3090 for model training and testing. The specific experimental environment is: Windows 10 system, PyTorch 1.12, OpenCV 4.5, NumPy 1.21, Matplotlib 3.5.

B. Comparative Experimental Results and Analysis

To verify the effectiveness of the proposed method, comparative tests are conducted on the DUT Anti-UAV and Anti-UAV target tracking datasets respectively. The results are shown below.

TABLE I. *PERFORMANCE COMPARISON OF VARIOUS METHODS ON THE DUT ANTI-UAV TRACKING SUBSET*

Method	Success ↑	Precision ↑	FPS ↑
FT	0.601	0.834	77.8
IT	0.583	0.797	76.5
RT	0.595	0.809	75.2
AMFT	0.616	0.932	72.7

From the experimental results of the visible light modality in Table 1, AMFT always surpasses traditional baselines. Its core value lies in breaking through the limitations of single trigger mechanisms in complex scenarios through multi-feature fusion, and realizing the coordinated optimization of robustness, real-time performance and generalization ability.

In terms of comprehensive accuracy, when AMFT faces challenges such as "out-of-view" and "occlusion" in visible light scenarios, it avoids the one-sidedness of single indicators such as fixed thresholds and geometric overlap through the collaborative judgment of motion continuity, background interference and appearance features. The comprehensive accuracy is stably ahead of the baselines. Even if the baselines have performance fluctuations due to relying on single features, AMFT still maintains stable accuracy, reflecting its adaptability to scene changes.

In terms of tracking stability and positioning accuracy, AMFT does not rely on fixed parameters or single-scene assumptions, and can dynamically adjust the trigger strategy according to dynamic changes such as fast target movement and occlusion. It optimizes the positioning deviation under visible light, and both tracking continuity and position accuracy are ahead of the baselines, confirming the effectiveness of the dynamic adaptation mechanism.

In terms of real-time performance, although multi-feature calculation has overhead, AMFT reduces invalid global detection by accurately judging the trigger timing, and maintains the real-time efficiency that meets the needs of UAV onboard deployment under visible light, forming a positive cycle of "accuracy improvement without sacrificing efficiency".

In conclusion, the core value of AMFT is to reconstruct the judgment logic of the trigger mechanism,

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solve the fundamental limitations of traditional methods in complex scenarios, and provide a solution with both practicality and expansibility for UAV detection-tracking collaboration.

V. CONCLUSIONS

Aiming at the core problems of rigid trigger mechanisms and single feature evaluation in vision-based Anti-UAV systems, this study proposes a "tracking-led, detection-on-demand triggering" collaboration framework centered on AMFT (Adaptive Multi-Feature Fusion Trigger Mechanism), aiming to improve the robustness and real-time performance of UAV tracking in dynamic scenarios. AMFT quantifies the tracking state through the "confidence-mobility-interference degree" three-dimensional evaluation system, and combines the sliding window dynamic threshold module to accurately adapt the scene complexity and trigger conditions, avoiding the defects of traditional fixed thresholds and single features. Experimental verification shows that in the visible light modality, its tracking stability and positioning accuracy are significantly better than baseline methods such as fixed thresholds and IoU triggers, and its real-time performance meets the deployment requirements.

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